

React.js Complete Course (Basic to Advanced)

Module 1: Introduction to React

Lesson 1

- What is React.js?
 - History of React
 - Why React?
 - Features of React
 - Advantages & Disadvantages
 - React vs JavaScript
 - React vs Angular
 - React vs Vue
 - Single Page Applications (SPA)
 - Virtual DOM
 - Real DOM
-

Lesson 2

Environment Setup

- Install Node.js
- Install VS Code
- Install Extensions
- Create React App
- Vite Introduction

Create Project using Vite

```
npm create vite@latest my-app
```

Install Dependencies

```
cd my-app  
npm install
```

Run Project

npm run dev

Module 2: React Basics

Lesson 3

- Folder Structure
- Components
- JSX
- Expressions in JSX
- Fragment

Example

```
function App(){  
  return <h1>Hello React</h1>;  
}
```

Lesson 4

Components

- Functional Components
- Reusable Components
- Nested Components
- Component Naming Rules

Projects

- Welcome Component
 - Student Component
-

Module 3: JSX

Lesson 5

Topics

- JSX Syntax
- JavaScript inside JSX
- Expressions
- Comments
- Attributes

Projects

- Student Card
 - Product Card
-

Module 4: Props

Lesson 6

Topics

- Props
- Passing Data
- Default Props
- Children Props

Projects

- Employee Card
 - Movie Card
-

Module 5: State

Lesson 7

Hooks

- useState()

Projects

- Counter
- Like Button
- Toggle Theme

Module 6: Events

Lesson 8

Events

- onClick
- onChange
- onSubmit
- onMouseEnter
- onKeyUp

Projects

- Button Counter
 - Keyboard Event Demo
-

Module 7: Conditional Rendering

Lesson 9

Topics

- if
- ternary operator
- &&
- switch

Projects

- Login System
 - User Dashboard
-

Module 8: Lists

Lesson 10

Topics

- `map()`
- `keys`
- Dynamic Rendering

Projects

- Student List
 - Product List
-

Module 9: Forms

Lesson 11

Topics

- Controlled Components
- Input Fields
- Textarea
- Select
- Checkbox
- Radio Button

Projects

- Registration Form
-

Module 10: CSS in React

Lesson 12

Topics

- CSS File
- Inline CSS
- CSS Modules
- Styled Components (Introduction)

Projects

- Responsive Navbar
-

Module 11: Hooks

Lesson 13

Hooks

- useState
- useEffect

Projects

- Digital Clock
 - API Loader
-

Lesson 14

Advanced Hooks

- useRef
- useMemo
- useCallback
- useReducer

Projects

- Stopwatch
 - Calculator
-

Module 12: React Router

Lesson 15

Install

npm install react-router-dom

Topics

- BrowserRouter
- Routes
- Route
- Link
- NavLink
- useNavigate
- URL Parameters

Projects

- Multi-page Website
-

Module 13: API Integration

Lesson 16

Topics

- Fetch API
- Axios

Install

npm install axios

Projects

- Weather App
 - Student API
-

Module 14: CRUD Operations

Lesson 17

Projects

- Student CRUD
- Employee CRUD
- Product CRUD

Module 15: Context API

Lesson 18

Topics

- createContext
- useContext
- Provider

Projects

- Theme Switcher
 - User Login
-

Module 16: Authentication

Lesson 19

Topics

- Login
- Logout
- Protected Routes
- JWT Token

Projects

- Login System
-

Module 17: State Management

Lesson 20

Topics

- Redux Toolkit
- Redux Store
- Actions
- Reducers

Install

npm install @reduxjs/toolkit react-redux

Projects

- Shopping Cart
-

Module 18: Custom Hooks

Lesson 21

Topics

- Creating Custom Hooks
- Reusable Logic

Projects

- Window Size Hook
-

Module 19: Performance Optimization

Lesson 22

Topics

- React.memo
- Lazy Loading
- Code Splitting
- Suspense

Projects

- Product Catalog

Module 20: Forms Validation

Lesson 23

Libraries

- React Hook Form
- Yup

Projects

- Student Registration
-

Module 21: File Upload

Lesson 24

Topics

- Image Upload
- Preview Image

Projects

- Profile Upload
-

Module 22: Charts

Lesson 25

Libraries

- Chart.js
- Recharts

Projects

- Sales Dashboard
-

Module 23: UI Libraries

Lesson 26

Topics

- Bootstrap
- Material UI
- Tailwind CSS (Introduction)

Projects

- Admin Dashboard
-

Module 24: Testing

Lesson 27

Tools

- Jest
- React Testing Library

Projects

- Component Testing
-

Module 25: Deployment

Lesson 28

Deploy To

- Vercel

- Netlify
- GitHub Pages

Projects

- Portfolio Website
-

Module 26: Best Practices

Lesson 29

Topics

- Folder Structure
 - Naming Conventions
 - Component Reusability
 - Performance
 - Error Boundaries
-

Module 27: Final Projects

Lesson 30

Build Complete React Applications

Projects

1. Student Management System
2. Employee Management System
3. Library Management System
4. Hospital Management System
5. Hotel Booking System
6. Portfolio Website
7. E-Commerce Website
8. Learning Management System (LMS)
9. Expense Tracker
10. Movie Search App
11. Weather Dashboard
12. Food Delivery App
13. Chat Application (Frontend)

14. Online Examination System
 15. CRM Dashboard
 16. Banking Dashboard
 17. Admin Panel
 18. Inventory Management
 19. Blog Website
 20. Company Website
-

30 Hands-on Practice Projects

1. Hello React
 2. Greeting Component
 3. Counter App
 4. Digital Clock
 5. Theme Switcher
 6. Student Card
 7. Product Card
 8. Employee List
 9. Todo List
 10. Calculator
 11. BMI Calculator
 12. Registration Form
 13. Login Form
 14. Stopwatch
 15. Weather App
 16. Notes App
 17. Student CRUD
 18. Employee CRUD
 19. Product CRUD
 20. Shopping Cart
 21. Quiz App
 22. Movie Search App
 23. Expense Tracker
 24. Library Management UI
 25. Chat UI
 26. Dashboard
 27. Portfolio Website
 28. Admin Panel
 29. Learning Management System
 30. Complete E-Commerce Website
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React.js Interview Questions (Basic to Advanced)

Beginner

1. What is React?
2. What are the features of React?
3. What is JSX?
4. What is a component?
5. Difference between functional and class components?
6. What are props?
7. What is state?
8. What is Virtual DOM?
9. What is the difference between state and props?
10. What is `useState()`?

Intermediate

11. What is `useEffect()`?
12. Explain React Router.
13. What is Context API?
14. What is Redux Toolkit?
15. What are controlled components?
16. What is lifting state up?
17. Explain React Hooks.
18. Difference between `useMemo()` and `useCallback()`?
19. What are custom hooks?
20. How do you fetch API data?

Advanced

21. What is reconciliation?
 22. Explain React Fiber.
 23. What is code splitting?
 24. What is lazy loading?
 25. What are Error Boundaries?
 26. How do you optimize React performance?
 27. Explain SSR vs CSR.
 28. What is hydration?
 29. What is React Strict Mode?
 30. What are React best practices?
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20 Real-World React Projects

1. Portfolio Website
 2. Student Management System
 3. Employee Management System
 4. Learning Management System (LMS)
 5. Hospital Management Dashboard
 6. E-Commerce Website
 7. Food Delivery App
 8. Banking Dashboard
 9. CRM Dashboard
 10. Inventory Management System
 11. Movie Search Application
 12. Weather Dashboard
 13. Expense Tracker
 14. Quiz Application
 15. Chat Application UI
 16. Admin Dashboard
 17. Library Management System
 18. Hotel Booking System
 19. Online Examination System
 20. Company Website
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React.js Cheat Sheet

Create Project

```
npm create vite@latest my-app  
cd my-app  
npm install  
npm run dev
```

Install Packages

```
npm install react-router-dom  
npm install axios  
npm install @reduxjs/toolkit react-redux  
npm install react-hook-form  
npm install yup
```

Common Hooks

```
useState()  
useEffect()  
useContext()
```

useRef()
useMemo()
useCallback()
useReducer()

Routing

```
<BrowserRouter>  
<Routes>  
<Route path="/" element={<Home />} />  
</Routes>  
</BrowserRouter>
```

API Call

fetch()
axios.get()

State Management

Context API
Redux Toolkit
